

NOVOLAZAREVSKAJA STATION, Antarctica

WMO: 895120

Lat: 70.777S

Lon: 11.821E

Elev: 119

StdP: 99.90

Time Zone: 0.00 (TRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-28.8	-27.0	-39.1	0.1	-26.4	-37.6	0.1	-25.7	52.4	-10.9	42.2	-12.3	1.5	80	1.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.6	3.7	-0.4	2.8	-1.0	2.1	-1.5	0.0	3.1	-0.7	2.3	-1.3	1.7	6.3	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
-3.4	2.9	0.5	-4.3	2.7	0.0	-5.1	2.5	-0.2	9.9	3.2	8.6	2.5	7.7	1.8	4.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 41.8	33.9	28.1	-32.4	5.9	2.5	1.9	-34.3	7.3	-35.7	8.4	-37.2	9.5	-39.0	10.9		
(5)			-32.4	1.2	2.2	1.3	-33.9	2.1	-35.2	2.9	-36.4	3.6	-38.0	4.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-10.3	-0.8	-3.5	-8.2	-11.8	-13.8	-15.0	-17.1	-17.2	-17.0	-12.3	-5.8	-1.2		
	DBStd	7.14	1.90	2.30	3.10	4.21	4.43	4.81	5.18	5.16	4.59	4.24	3.10	2.03		
	HDD10.0	7423	333	378	564	656	739	749	841	842	809	692	474	346		
	HDD18.3	10465	592	611	822	906	997	999	1099	1101	1059	951	724	604		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	11.7	7.9	11.0	11.7	14.5	13.5	13.7	13.5	12.9	10.5	11.5	10.8	8.6		
(15) Precipitation	PrecAvg	249	18	20	17	16	24	23	22	24	16	23	26	20		
	PrecMax	326	55	44	37	37	53	64	60	69	35	46	69	49		
	PrecMin	192	4	5	5	1	5	5	4	7	2	5	1	3		
	PrecStd	32	10	10	7	8	11	12	12	15	9	9	17	10		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	2.8	-0.3	-1.9	-4.4	-4.5	-5.1	-5.8	-6.1	-2.1	2.1	4.7		
		MCWB	0.9	-1.0	-3.4	-4.4	-7.1	-6.7	-7.6	-8.0	-8.5	-5.1	-1.7	0.2		
	2%	DB	3.9	1.7	-1.6	-3.9	-5.5	-6.3	-7.5	-7.8	-7.6	-3.8	0.7	3.6		
		MCWB	-0.1	-1.9	-4.5	-6.6	-7.8	-8.5	-9.3	-9.5	-9.9	-6.6	-3.0	-0.6		
	5%	DB	3.0	0.8	-2.6	-4.9	-6.7	-7.3	-8.7	-9.1	-9.0	-5.3	-0.4	2.8		
		MCWB	-0.7	-2.5	-5.3	-7.3	-8.7	-9.2	-10.2	-10.6	-11.0	-7.9	-3.8	-1.1		
	10%	DB	2.3	0.0	-3.7	-6.1	-7.9	-8.4	-10.1	-10.4	-10.4	-6.5	-1.5	2.1		
		MCWB	-1.2	-3.2	-6.3	-8.1	-9.7	-10.1	-11.6	-11.8	-12.3	-8.7	-4.6	-1.6		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	-0.4	-2.9	-4.3	-6.7	-5.8	-7.4	-7.5	-8.2	-4.8	-1.2	0.9		
		MCDB	4.4	2.2	-1.1	-2.2	-5.2	-4.8	-5.9	-6.2	-6.6	-2.6	1.6	4.0		
	2%	WB	0.2	-1.4	-4.2	-6.2	-7.5	-7.9	-9.0	-9.1	-9.7	-6.5	-2.8	-0.3		
		MCDB	3.4	1.2	-1.9	-4.3	-5.9	-6.6	-7.9	-8.0	-7.8	-4.3	0.3	2.9		
	5%	WB	-0.4	-2.3	-5.2	-7.2	-8.5	-9.1	-10.2	-10.5	-11.0	-7.6	-3.6	-0.8		
		MCDB	2.6	0.4	-2.8	-5.2	-6.9	-7.5	-8.8	-9.3	-9.0	-5.5	-0.7	2.3		
	10%	WB	-1.0	-3.0	-6.2	-8.1	-9.6	-10.1	-11.5	-11.7	-12.3	-8.6	-4.4	-1.3		
		MCDB	1.9	-0.4	-3.8	-6.1	-8.0	-8.4	-10.1	-10.4	-10.5	-6.6	-1.7	1.7		
(35) Mean Daily Temperature Range	MDBR		3.6	3.5	3.6	3.9	4.1	4.2	4.4	4.6	4.7	4.2	3.8	3.7		
	5% DB	MCDBR	4.4	4.2	3.7	4.2	3.7	4.3	4.4	4.5	5.2	4.9	4.4	4.4		
		MCWBR	3.2	3.4	3.2	3.4	3.1	3.7	3.7	3.9	4.4	4.1	3.4	3.1		
	5% WB	MCDBR	4.0	3.9	3.6	3.9	3.5	4.0	4.2	4.2	5.0	4.5	4.0	4.1		
		MCWBR	3.1	3.3	3.2	3.4	3.1	3.7	3.7	3.8	4.3	3.9	3.3	3.0		
(40) Clear-Sky Solar Irradiance	taub		0.243	0.229	0.199	0.143	N/A	N/A	N/A	0.139	0.180	0.202	0.226	0.244		
	taud		2.194	2.220	2.219	2.016	N/A	N/A	N/A	1.976	2.158	2.171	2.173	2.163		
	Ebn at Noon		997	948	864	622	0	0	0	608	876	980	1015	1013		
	Edh at Noon		117	96	67	31	0	0	0	31	68	99	118	126		
(44) All-Sky Solar Radiation	RadAvg		8.55	5.64	2.75	0.69	0.00	0.00	0.00	0.29	1.88	4.59	7.78	9.66		
	RadStd		0.18	0.20	0.07	0.03	0.00	0.00	0.00	0.02	0.08	0.16	0.24	0.20		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page